Problem 10. Write the solution set of

$$\begin{cases} 2x_2 - x_3 + 3x_5 = 0 \\ -x_1 + 2x_2 + 4x_3 - x_4 = 0 \\ x_4 + x_5 = 0 \end{cases}$$

as the image of a matrix.

Solution. To express the solution set as the image of a matrix, we first carry out the usual process of solving the system. The augmented matrix is as follows:

$$\left[\begin{array}{ccccc|c} 0 & 2 & -1 & 0 & 3 & 0 \\ -1 & 2 & 4 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{array}\right]$$

Applying row operations, we bring this to reduced row echelon form:

$$\rightsquigarrow \left[\begin{array}{ccccc|c} 1 & -2 & -4 & 1 & 0 & 0 \\ 0 & 2 & -1 & 0 & 3 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{array}\right] \rightsquigarrow \left[\begin{array}{ccccc|c} 1 & 0 & -5 & 1 & 3 & 0 \\ 0 & 2 & -1 & 0 & 3 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{array}\right] \rightsquigarrow \left[\begin{array}{ccccc|c} 1 & 0 & -5 & 1 & 3 & 0 \\ 0 & 1 & -1/2 & 0 & 3/2 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{array}\right]$$

$$\rightsquigarrow \left[\begin{array}{ccccc|c} 1 & 0 & -5 & 0 & 2 & 0 \\ 0 & 1 & -1/2 & 0 & 3/2 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{array}\right]$$

The free variables are x_3 and x_5, and we have

$$x_1 = 5x_3 - 2x_5$$

$$x_2 = \frac{1}{2}x_3 - \frac{3}{2}x_5$$

$$x_4 = -x_5$$

Thus our general solution is

$$\begin{bmatrix} 5x_3 - 2x_5 \\ x_3/2 - 3x_5/2 \\ x_3 \\ -x_5 \\ x_5 \end{bmatrix} = x_3 \begin{bmatrix} 5 \\ 1/2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -2 \\ -3/2 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

The solution set is thus

$$\operatorname{span} \left\{ \begin{bmatrix} 5 \\ 1/2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ -3/2 \\ 0 \\ -1 \\ 1 \end{bmatrix} \right\} = \operatorname{im} \begin{bmatrix} 5 & -2 \\ 1/2 & -3/2 \\ 1 & 0 \\ 0 & -1 \\ 0 & 1 \end{bmatrix}$$

Problem 11. Find a system of equations which defines the linear subspace

$$V = \operatorname{span} \left\{ \begin{bmatrix} 2 \\ 1 \\ 4 \\ -2 \end{bmatrix}, \begin{bmatrix} -4 \\ 1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Use these equations to determine if the following vectors are contained in V:

$$\begin{bmatrix}1\\1\\1\\1\end{bmatrix}, \begin{bmatrix}8\\1\\8\\-5\end{bmatrix}, \begin{bmatrix}-6\\0\\-4\\3\end{bmatrix}, \begin{bmatrix}2\\-1\\1\\-3\end{bmatrix}$$

Solution. A general vector

$$\begin{bmatrix}x_1\\x_2\\x_3\\x_4\end{bmatrix}$$

is in V if and only if the linear system with augmented matrix

$$\left[\begin{array}{cc|c}2 & -4 & x_1\\1 & 1 & x_2\\4 & 0 & x_3\\-2 & 1 & x_4\end{array}\right]$$

is solvable. We simplify the system using row operations as follows:

$$\rightsquigarrow \left[\begin{array}{cc|c}2 & -4 & x_1\\1 & 1 & x_2\\1 & 0 & x_3/4\\-2 & 1 & x_4\end{array}\right] \rightsquigarrow \left[\begin{array}{cc|c}0 & -4 & x_1 - x_3/2\\0 & 1 & x_2 - x_3/4\\1 & 0 & x_3/4\\0 & 1 & x_4 + x_3/2\end{array}\right] \rightsquigarrow \left[\begin{array}{cc|c}0 & 0 & x_1 + 4x_2 - 3x_3/2\\0 & 1 & x_2 - x_3/4\\1 & 0 & x_3/4\\0 & 0 & x_4 + 3x_3/4 - x_2\end{array}\right]$$

The condition for the system to be consistent is that the quantities opposite the zero rows are zero, that is:

$$\begin{cases} x_1 + 4x_2 - 3x_3/2 = 0\\ x_4 + 3x_3/4 - x_2 = 0\end{cases}$$

Now to check if the given vectors are in V, we check if they satisfy these equations. The first vector

$$\begin{bmatrix}1\\1\\1\\1\end{bmatrix}$$

satisfies neither equation, so is not in V. The second vector

$$\begin{bmatrix}8\\1\\8\\-5\end{bmatrix}$$

satisfies both, so is in V. The third vector

$$\begin{bmatrix}-6\\0\\-4\\3\end{bmatrix}$$

also satisfies both, so is in V. The fourth vector

$$\begin{bmatrix}2\\-1\\1\\-3\end{bmatrix}$$

satisfies neither equation, so is not in V.

Problem 12. Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be a linear transformation. Prove that the *fixed point set* of T,

$$\{v \in \mathbb{R}^n \mid T(v) = v\}$$

is a linear subspace of $\mathbb{R}^n$.

Solution. Let $U = \{v \in \mathbb{R}^n \mid T(v) = v\}$. To check that U is a linear subspace, we need to check three things:

1. $0 \in U$
2. If $u, v \in U$, then $u + v \in U$
3. If $u \in U$ and $c \in \mathbb{R}$, then $cu \in U$.

To check (1), we need to check that $T(0) = 0$. This is true because T is linear.

To check (2), assume that $u, v \in U$. This means that $T(u) = u$ and $T(v) = v$. Now

$$T(u + v) = T(u) + T(v) = u + v$$

using the linearity of T. This implies that $u + v \in U$.

To check (3) assume that $u \in U$ and $c \in \mathbb{R}$. Then

$$T(cu) = cT(u) = cu$$

again using the linearity of T for the first equation. This implies that $cu \in U$.

Having checked (1), (2), and (3), we can conclude that U is a linear subspace.

Solution (Alternative). If T corresponds to a matrix A, then $T(v) = v$ if and only if $0 = Av - v = (A - I_n)v$, so $U = \ker(A - I_n)$ is a subspace because the kernel of any matrix is a subspace.

Problem 13. Prove that if v_1, v_2, v_3, v_4 is a linearly independent set of vectors, then so is v_1, v_3.

Solution. Suppose that v_1, v_2, v_3, v_4 is a linearly independent set of vectors. In order to show that v_1, v_3 is a linearly independent set of vectors, we need to check that if $av_1 + bv_3 = 0$, then it must be that $a = b = 0$. If we have $av_1 + bv_3 = 0$, then we can rewrite this as a linear combination of v_1, v_2, v_3, v_4 as follows:

$$0 = av_1 + bv_3 = av_1 + 0v_2 + bv_3 + 0v_4$$

Now, because v_1, v_2, v_3, v_4 is a linearly independent set, all the coefficients in this must be 0, and in particular $c_1 = c_3 = 0$.

Problem 14. Suppose that $T : \mathbb{R}^n \to \mathbb{R}^n$ is a linear transformation. Prove that $S(v) = \frac{1}{2}(v + T(v))$ is a linear transformation. If T corresponds to a matrix A, what is the matrix corresponding to S?

Solution. Suppose that T is a linear transformation. If the function $S : \mathbb{R}^n \to \mathbb{R}^n$ is defined by

$$S(v) = \frac{1}{2}(v + T(v))$$

then to check that S is a linear transformation, we need to check two things:

1. $S(u + v) = S(u) + S(v)$ for vectors u, v
2. $S(cv) = cS(v)$ for a vector v and a scalar c.

To check (1), we apply linearity of T to the expression for $S(u + v)$:

$$S(u + v) = \frac{1}{2}(u + v + T(u + v)) = \frac{1}{2}(u + v + T(u) + T(v))$$

We then rearrange this expression to get

$$\frac{1}{2}(u + T(u)) + \frac{1}{2}(v + T(v)) = S(u) + S(v)$$

as required.

Checking (2) is similar. We apply the linearity of T to the expression for $S(cu)$:

$$S(cu) = \frac{1}{2}(cu + T(cu)) = \frac{1}{2}(cu + cT(u))$$

Now we can factor out c, which yields

$$\frac{c}{2}(u + T(u)) = cS(u)$$

as required.

Now to find the matrix, the fact that T corresponds to S means that $T(v) = Av$ for all vectors v. We can then write

$$S(v) = \frac{1}{2}(v + T(v)) = \frac{1}{2}(v + Av) = \frac{1}{2}(I_n v + Av) = \frac{1}{2}(I_n + A)v$$

Therefore, S corresponds to the matrix $\frac{1}{2}(I_n + A)$.

Solution (Alternative)*.* If T corresponds to A, then we can write

$$S(v) = \frac{1}{2}(v + T(v)) = \frac{1}{2}(I_n v + Av) = \frac{1}{2}(I_n + A)v$$

Because S is given by multiplication by the matrix $\frac{1}{2}(I_n + A)$, it is a linear transformation.